Round 2
ADVANCED TECHNICAL ROUND

1) Array Rotation

Rotate an array by k positions to the right without using extra space.

Input:

```
n = 7
arr = [1, 2, 3, 4, 5, 6, 7]
k = 3
```

Output:

```
[5, 6, 7, 1, 2, 3, 4]
```

2) Coin Change (DP)

Given coins of different denominations and a target amount, find the minimum number of coins needed. If not possible, return -1.

Input:

```
coins = [1, 2, 5]
amount = 11
```

Output:

3 (11 = 5 + 5 + 1)

1. Study the flow chart give below and the questions that follow

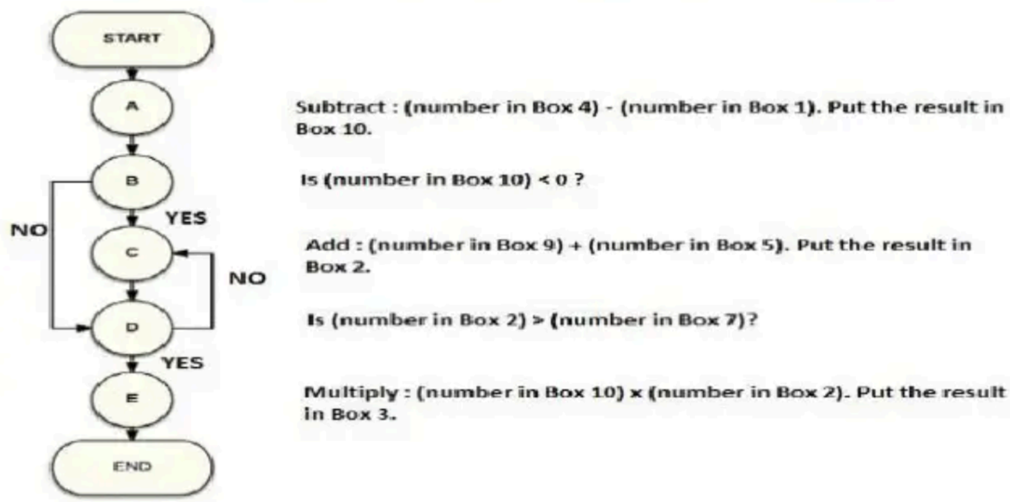
Question: At the end of the flow chart the number placed in which of the following boxes will remain unchanged

Box No.

1	2	3	4	5	6	7	8	9	10
13	20	7	12	10	2	5	1	0	18

Box No.

1	2	3	4	5	6	7	8	9	10
13	20	7	12	10	2	5	1	0	18

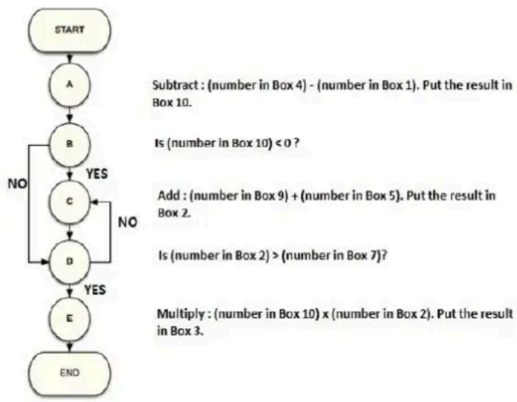


A. Box 3 **B.** Box 10 **C.** Box 2 **D.** Box 7

2. Is the following statement True or False
Statement:
If the condition in Step C updates the value in Box 3 instead of Box 2, then the flow chart will enter into infinite loop

Box No.

1	2	3	4	5	6	7	8	9	10
13	20	7	12	10	2	5	1	0	18

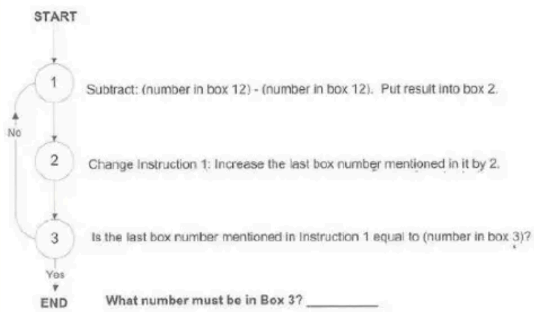


A. TRUE B. FALSE C. Cannot be determined

3. Study the flow chart give below and the questions that follow.
The purpose of the following chart is to put a zero in each of the boxes 2,4 and 6. In order to accomplish exactly this - no more and no less - Which number must be in box 3?

Box No.	1	2	3	4	5	6	7	8	9	10	11	12
	7	6		1	6	10	4	3	2	14	8	9

Box No.	1	2	3	4	5	6	7	8	9	10	11	12
	7	6		1	6	10	4	3	2	14	8	9



A. 0 **B.** 6 **C.** 10 **D.** 3 **E.** 8